Dimension Importance Estimation via Genetic Algorithms*

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Abstract

Dense retrieval has transformed neural Information Retrieval (IR) by representing queries and documents in continuous vector spaces and ranking documents by semantic similarity rather than lexical overlap. However, not all embedding dimensions contribute equally to retrieval effectiveness. Prior work shows that some dimensions can be removed with little impact, while others are crucial or even detrimental to performance. Dimension IMportance Estimators (DIMEs) exploit this observation by identifying subsets of dimensions that outperform the full representation. Previous approaches simplify this task by restricting the search to linear subspaces and evaluating dimensions independently. We retain the linear-subspace assumption but instead model dimensions jointly, using a genetic algorithm to search for effective combinations.

Experiments show substantial gains: with query-specific dimension selection, state-of-the-art dense retrievers achieve up to 0.981 nDCG@10 and 0.831 AP. Moreover, subsets learned on one dataset transfer to others without reducing effectiveness, suggesting that useful dimensional structure can generalize across other datasets without loss of performance.

1. Introduction

Recent advances in Large Language Models (LLMs) have enabled Information Retrieval (IR) models to move from keyword-based to semantic search, improving ranking effectiveness in handling language variations, synonyms, and contextual ambiguities [2]. Modern retrieval systems exploit these models to capture the semantic meaning of documents and queries. In particular, dense IR systems use contextualized embeddings [3] to represent queries and documents as dense vectors in a shared, lower-dimensional latent space, where each dimension encodes a latent feature. However, dense representations are less interpretable than lexical ones: we do not know what individual dimensions represent, why they take certain values, or whether they are necessary [4]. Recent work shows that dimensions do not contribute equally to ranking quality: some are useful, while others can harm performance [5, 6]. Building on this observation, Faggioli et al. [5] introduced the *manifold clustering hypothesis*, showing that retrieval effectiveness can be improved by projecting queries and documents onto query-specific subspaces of the original latent space. To identify such subspaces, Faggioli et al. developed heuristics called Dimension IMportance Estimators (DIMEs). To make the problem tractable, DIMEs rely on two simplifying assumptions: (i) restricting the search to *linear subspaces*, and (ii) assuming *dimension independence*. While these assumptions reduce complexity, the latter ignores interactions between dimensions that may be important for characterizing the embedding space.

This paper further investigates the manifold clustering hypothesis by asking whether more effective query-specific DIMEs can be found by relaxing the independence assumption. We also study whether the gains induced by DIME-based shifts in reference space persist across queries and whether dimensionality reduction strategies transfer across collections. We formalize these objectives through the following research questions:

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RQ1 **Effective dimensionality reduction**. To what extent can query-specific dimension selection improve performance?

RQ2 **Generalization on multiple datasets**. Can we learn a robust subset of dimensions that enables a model to generalize across different datasets?

This paper is primarily theoretical in nature. However, given the analytical intractability of studying the manifold clustering hypothesis, we adopt an empirical heuristic approach. To explore subspaces without assuming independent dimensions, we use genetic algorithms (GAs) to approximate optimal subsets of dimensions. These are guided by cost functions defined using relevance judgments from ground truth pools, placing our study in an oracle setting with ideal, noise-free feedback. Our genetic approach, applied on a query-per-query basis, selects subspaces leading to substantial performance gains, improving nDCG@10 up to +113% and AP up to +191%. Finally, we demonstrate that dimension subsets learnt on one dataset can be transferred to other datasets without degrading effectiveness, indicating promising generalization capabilities.

2. Methodology

A bi-encoder dense retrieval model maps a query q and corpus $\mathcal{C} = \{d_1, ..., d_n\}$ into a latent space $\mathbb{R}^h$. Given representations $\mathbf{q}, \mathbf{d} \in \mathbb{R}^h$, the retrieval score is their dot product $\mathbf{q} \cdot \mathbf{d}$. The ranked list $\mathcal{R}_q = \{d_1, ..., d_k\}$ contains the top-k documents by score, and, given relevance labels $\mathcal{L}$, its effectiveness is measured by $M(\mathcal{R}_q; \mathcal{L})$. Faggioli et al. [5] observed that there may exist a linear subspace[1] $\mathbb{R}^{|\delta|}$ defined by dimensions $\delta \subset \{1, ..., h\}$, with $|\delta| < h$, in which retrieval is more effective. Denoting by $\mathcal{R}q, \delta$ the ranking obtained in this subspace, we have $M(\mathcal{R}q; \mathcal{L}) < M(\mathcal{R}q, \delta; \mathcal{L})$. We denote by $(\mathbf{q} \cdot \mathbf{d})\delta$ the corresponding dot product in $\mathbb{R}^{|\delta|}$. Finding the optimal δ is intractable because $\mathbb{R}^h$ has 2^h possible linear subspaces. Hence, Faggioli et al. propose Dimension IMportance Estimators (DIMEs), $u : (1, ..., h; \mathbf{q}, \theta) \mapsto \mathbb{R}$, which assigns an importance score to each dimension given its index, the query representation $\mathbf{q}$, and optional inputs θ. The DIME task can be formulated as finding the subset of dimensions that maximizes retrieval effectiveness. In this paper, we use GAs to identify optimal subsets for individual queries. Unlike the heuristic approaches in [5, 6, 7, 8, 9], our evolutionary framework accounts for dependencies between dimensions while balancing exploration and exploitation. Each individual represents a subset of dimensions as a binary string, where 1 indicates that the i-th dimension is retained and 0 otherwise. The population is randomly initialized, and at each generation, fitness-based selection favors subsets yielding higher ranking quality while preserving diversity. New individuals are produced by recombining selected parents. We also employ elitism, ensuring that the best solution of each generation is carried unchanged to the next. The search is guided through a fitness function. We define two fitness functions to answer the two research questions.

Effective Dimensionality Reduction RQ1 focuses on determining the largest improvement achievable by a query-specific DIME when the dimension-independence assumption is relaxed. To approximate the optimal subset, we define the following objective function, which guides the GA for each query q using the relevance judgments $\mathcal{L}$:

$$\max_{\delta_q} \frac{1}{|\mathcal{M}|} \sum_{M \in \mathcal{M}} M(\mathcal{R}_{q,\delta_q}; \mathcal{L}) - \frac{|\delta_q|}{h}$$

where δ_q represents the subset of dimensions selected for query q (i.e., an individual in the GA) and $\mathcal{M}$ is the set of evaluation measures to maximize. This objective function seeks to identify the subset of dimensions that maximizes the evaluation measures while simultaneously minimizing the number of dimensions considered. In our experiments we set $\mathcal{M} = \{\text{nDCG@10}, \text{AP}\}$. This approach does not

[1] Faggioli et al. postulate that the subspace can be an arbitrary manifold, but focus only on independent linear subspaces, i.e., subspaces constructed by removing some dimensions.

require explicitly specifying the number of retained dimensions, as the GA automatically selects it for each query according to the objective function.

Generalization on Multiple Datasets We aim to identify a robust subset of dimensions that preserves retrieval gains across datasets without relying on dataset-specific relevance labels. Specifically, we select dimensions on one training dataset and evaluate whether the improvement transfers to other test datasets. Successful transfer would suggest systematic noise in the embedding space, with some dimensions consistently harming retrieval effectiveness. We fix one dataset as training data, select the retained dimensions through our optimization problem, and evaluate them on the remaining datasets. Unlike the previous setting, this setting uses proper train/test splits, providing both an analysis of latent dimension content and a methodology applicable to real-world scenarios. Let $\mathcal{L}_q^+$ and $\mathcal{L}_q^-$ denote the relevant and non-relevant documents for query q in the training dataset. We define:

$$\max_{\delta} \sum_{q \in Q} \left(\frac{1}{|\mathcal{L}_q^+|} \sum_{d^+ \in \mathcal{L}_q^+} (\mathbf{q} \cdot \mathbf{d}^+)_\delta - \frac{1}{|\mathcal{L}_q^-|} \sum_{d^- \in \mathcal{L}_q^-} (\mathbf{q} \cdot \mathbf{d}^-)_\delta \right)$$

The objective identifies dimensions that increase similarity to relevant documents while reducing similarity to non-relevant ones. Unlike the previous objectives, it does not optimize nDCG@10 or AP directly, but instead maximizes the separation between relevant and non-relevant documents. This is intended to identify dimension subsets with stronger generalization across datasets.

3. Experimental Assessment

Experimental Settings We empirically evaluate our research questions on three 768-dimensional dense retrieval models: Contriever [10], TAS-B [11], and Dragon [12]. All models are fine-tuned on MS-MARCO and publicly available on HuggingFace. We consider TREC Deep Learning '19 (DL 19) [13], TREC Deep Learning '20 (DL 20) [14], and TREC Robust '04 (RB 04) [15]. Our GA uses uniform crossover ($P_{crossover} = 0.5$), bit-flip mutation ($P_{mutation} = \frac{2}{768}$ per bit), and tournament selection of size 3. The randomly initialized population contains 100 individuals.

Table 1
Retrieval performance using the best query-specific subset of dimensions found by the GA compared to the baseline using all dimensions. * indicates statistically significant improvements.

		Genetic algorithm			Baseline		
Dataset	Model	nDCG@10	AP	Avg N° dims.	nDCG@10	AP	Avg N° dims.
DL 19	TAS-B	**0.975***	**0.813***	55	0.717	0.481	768
	Contriever	**0.964***	**0.811***	58	0.674	0.493	
	Dragon	**0.981***	**0.831***	56	0.740	0.522	
DL 20	TAS-B	**0.972***	**0.800***	57	0.684	0.478	768
	Contriever	**0.972***	**0.803***	55	0.672	0.482	
	Dragon	**0.976***	**0.824***	59	0.718	0.509	
RB 04	TAS-B	**0.953***	**0.619***	55	0.447	0.213	768
	Contriever	**0.960***	**0.659***	55	0.465	0.239	
	Dragon	**0.948***	**0.633***	58	0.461	0.228	

Effective Dimensionality Reduction Our first investigation concerns the maximum performance achievable through the selection performed by the GA of an effective query-specific subset of dimensions using the oracle optimization function. With our settings the GA examines for each query at most $500 \cdot 100 = 50,000$ possible subsets of dimensions in the $2^{768} \approx 1.55 \cdot 10^{231}$ possible subsets. Table 1 reports the oracle experiment results. Despite exploring only a limited portion of the solution space, the GA identifies dimension subsets that substantially improve retrieval effectiveness. On average, selecting about 8% of dimensions per query yields gains of up to 113% in nDCG@10 and 191% in AP on RB 04, with nDCG@10 approaching 1 and AP reaching ≈ 0.8. This leads to two observations:

- Fewer than 8% of dimensions contribute positively to ranking quality, while the remaining 92% are irrelevant or detrimental, suggesting that current training strategies may inject substantial noise into query representations.
- Although this setting is oracular and cannot be directly applied in practice, the gains are substantial, more than doubling effectiveness on RB 04. Future work could approximate these gains by identifying high-impact dimensions using proxy signals or noisy relevance judgments rather than curated ground truth.

Generalization on Multiple Datasets The previous experiment showed the effectiveness of the dimension selection at a query level. We now turn to generalizing across datasets, moving away from the oracle setting and considering a practical machine learning setting, with separate training and test sets. In this experiment, we consider DL 19 as the training dataset to find, with our GA and the second optimization function, a single good subset of dimensions, and we test whether the identified subset of dimensions generalizes well across the remaining datasets.

Table 2
Answering RQ3: retrieval performance on the target test dataset using the best subset of dimensions found by the GA using DL 19 as training dataset compared to the baseline using all dimensions.

		Genetic algorithm			Baseline		
Model	Dataset	nDCG@10	AP	N° dims.	nDCG@10	AP	N° dims.
TAS-B	DL 19 (Train)	**0.734**	**0.507***	674	0.717	0.481	768
	DL 20 (Test)	0.682	**0.479**		**0.684**	0.478	
	RB 04 (Test)	0.439	0.209*		**0.447**	**0.213**	
Contriever	DL 19 (Train)	**0.683**	**0.501**	668	0.674	0.493	768
	DL 20 (Test)	0.671	0.477		**0.672**	**0.482**	
	RB 04 (Test)	0.462	0.236*		**0.465**	**0.239**	
Dragon	DL 19 (Train)	**0.748**	0.517	671	0.740	**0.522**	768
	DL 20 (Test)	**0.735**	0.508		0.718	**0.509**	
	RB 04 (Test)	0.454	0.223*		**0.461**	**0.228**	

Table 2 shows the results of this experiment. The selected subset of dimensions enables dense retrieval models to generalize effectively across different datasets. By learning the set of dimensions on DL 19, this clearly induces a performance improvement. The optimal number of dimensions ranges from 668 (Contriever) to 674 (TAS-B). By reducing the dimensions by around 13% to 12%, we still maintain a good separation between the scores assigned to relevant and non-relevant documents. If we use the same set of dimensions to project the representation space also for DL 20 and RB 04, we notice a minor deflation of effectiveness. This decrease occurs in the third decimal place, indicating a negligible change in terms of ranking. In other terms, while we have saved more than 12% of the resources—both in terms of space on the disk and in the number of required operations to compute the dot product—it is highly unlikely that the user will perceive this change when looking at the ranked list of documents.

4. Conclusion and Future Work

In this paper, we investigated how strategically selecting dimensions from dense retrieval representations can improve ranking performance. We provided novel empirical evidence supporting the manifold clustering hypothesis by identifying effective subsets of dimensions for each query using DIMEs combined with GAs. Our experiments showed substantial improvements in nDCG@10 and AP across multiple models and datasets. We further showed that robust dimension subsets can generalize to unseen datasets without compromising effectiveness. Although these findings are mainly theoretical, as they largely rely on an oracle setting, they show that dense representations often contain noise or redundancy that can harm ranking quality. Selecting a smaller but informative subset of dimensions can therefore preserve effectiveness while also yielding substantial performance gains.

Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

GenAI disclosure

During the preparation of this work, the authors used Grammarly and ChatGPT to perform grammar and spelling checks, paraphrase, and reword under the full supervision of the authors. The authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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